Chapter 4 • Factorization and Algebraic Manipulation

Solved Review Exercise 4

MCQs

1. The factorization of 12x+36 is:
a) 12(x+3)
b) 12(3x)
c) 12(3x+1)
d) x(12+36x)
Ans: a
The common factor is 12.

2. The factors of $4x^2-12x+9$ are:
a) $(2x+3)^2$
b) $(2x-3)^2$
c) $(2x-3)(2x+3)$
d) $(2+3x)(2-3x)^2$
Ans: b
It is the perfect square $(2x-3)^2$.

3. The HCF of a^3b^3 and ab^2 is:
a) a^3b^3
b) ab^2
c) a^4b^5
d) a^2b
Ans: b
Use the lower power of each variable.

4. The LCM of $16x^2$, 4x and 30xy is:
a) $480x^3y$
b) 240xy
c) $240x^2y$
d) $120x^4y$
Ans: c
LCM of coefficients is 240; use x^2 and y.

5. Product of LCM and HCF equals the ___ of two polynomials.
a) sum
b) difference
c) product
d) quotient
Ans: c
For two polynomials, LCM×HCF=product.

6. The square root of x^2-6x+9 is:
a) $\pm(x-3)$
b) $\pm(x+3)$
c) $x-3$
d) $x+3$
Ans: a
$x^2-6x+9=(x-3)^2$.

7. The LCM of $(a-b)^2$ and $(a-b)^4$ is:
a) $(a-b)^2$
b) $(a-b)^3$
c) $(a-b)^4$
d) $(a-b)^6$
Ans: c
Use the greater exponent, 4.

8. Factorization of x^3+3x^2+3x+1 is:
a) $(x+1)^3$
b) $(x-1)^3$
c) $(x+1)(x^2+x+1)$
d) $(x-1)(x^2-x+1)$
Ans: a
This is the perfect-cube expansion of $(x+1)^3$.

9. A cubic polynomial has degree:
a) 1
b) 2
c) 3

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d) 4
Ans: c
Cubic means degree 3.

10. One factor of x^3-27 is:
a) $x-3$
b) $x+3$
c) x^2-3x+9
d) Both a and c
Ans: a
$x^3-27=(x-3)(x^2+3x+9)$.

2. Factorize all nine expressions.

(i) $4x^3+18x^2-12x=2x(2x^2+9x-6)$.
(ii) $x^3+64y^3=(x+4y)(x^2-4xy+16y^2)$.
(iii) $x^3y^3-8=(xy-2)(x^2y^2+2xy+4)$.
(iv) $-x^2-23x-60 = -(x^2+23x+60)$
$= -(x+3)(x+20)$.
(v) $2x^2+7x+3 = 2x^2+6x+x+3$
$= (2x+1)(x+3)$.
(vi) $x^4+64 = (x^2+8)^2-(4x)^2$
$= (x^2+4x+8)(x^2-4x+8)$.
(vii) $x^4+2x^2+9 = (x^2+3)^2-(2x)^2$
$= (x^2+2x+3)(x^2-2x+3)$.
(viii) $(x+3)(x+4)(x+5)(x+6)-360$: pair to obtain $(x^2+9x+18)(x^2+9x+20)-360$. Let $u=x^2+9x+19$: $(u-1)(u+1)-360=u^2-361=(u-19)(u+19)=x(x+9)(x^2+9x+38)$.
(ix) $(x^2+6x+3)(x^2+6x-9)+36$: let $u=x^2+6x-3$, then $(u+6)(u-6)+36=u^2=(x^2+6x-3)^2$.

3(i). Find LCM and HCF of $4x^3+12x^2$ and $8x^2+16x$.

$4x^3+12x^2=4x^2(x+3)$
$8x^2+16x=8x(x+2)$.
HCF$=4x$
LCM uses 8, x^2, $(x+2)$, $(x+3)$.
LCM$= 8x^2(x+2)(x+3)$, HCF$=4x$. The printed LCM has an extra x.

3(ii). Find LCM and HCF of x^3+3x^2-4x and x^2-4x+3.

Factors: $x(x+4)(x-1)$ and $(x-1)(x-3)$. LCM$=x(x+4)(x-1)(x-3)$; HCF$=x-1$.

3(iii). Find LCM and HCF of $x^2+8x+16$ and x^2-16.

$(x+4)^2$ and $(x-4)(x+4)$: LCM$=(x-4)(x+4)^2$; HCF$=x+4$.

3(iv). Find LCM and HCF of x^3-9x and x^2-x-6.

$x(x-3)(x+3)$ and $(x-3)(x+2)$: LCM$=x(x+2)(x^2-9)$; HCF$=x-3$.

4. Find the square root of $16x^4+8x^2+1$ by factorization and division.

$16x^4+8x^2+1 = (4x^2)^2+2(4x^2)(1)+1^2$
$= (4x^2+1)^2$.
The division algorithm gives the same square-root expression with zero final remainder.
$\pm(4x^2+1)$.

5. C(x)=$x^2-8x+15$; find the repayment values used by the textbook model.

C(x) = $(x-3)(x-5)$, so C(x)
$= 0$ at x
$= 3$ or x
$= 5$.
3 years or 5 years under the textbook's zero-cost interpretation.

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